

TEMPERATURE AND HUMIDITY RASPBERRY PI PROJECT

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INTRO

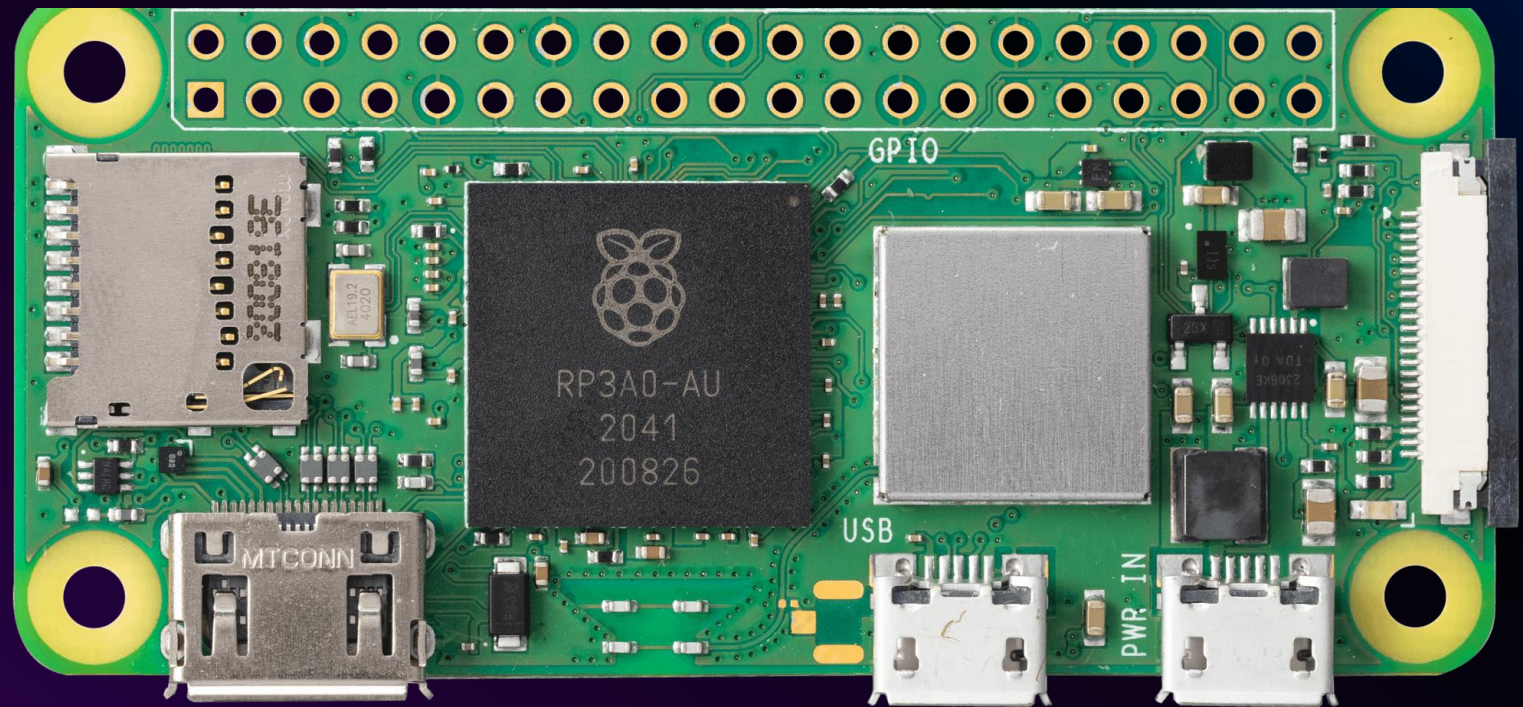
For our project we made a Temperature and Humidity sensor integrated with an Adafruit BME280 sensor. The sensor monitors the environment around itself and produces a live reading.



MATERIALS USED:

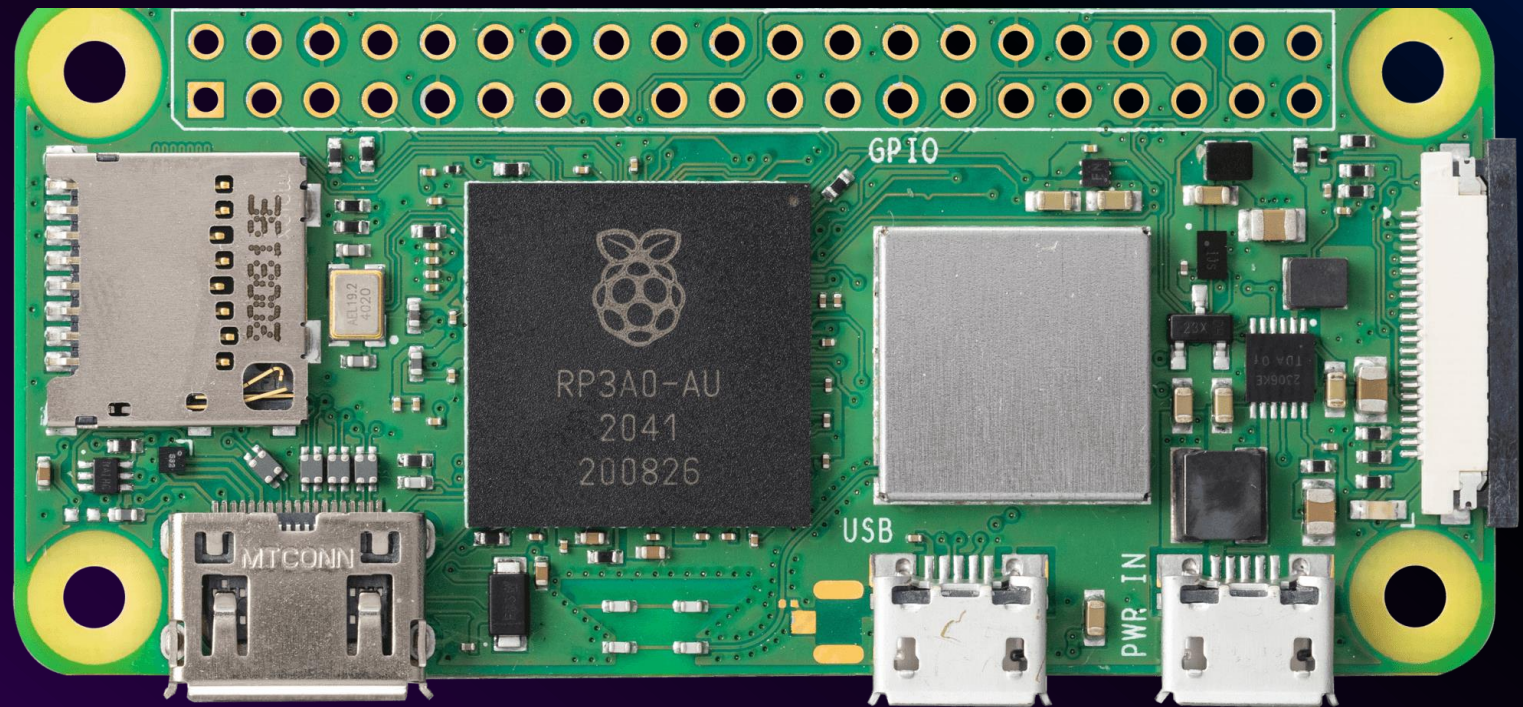
RASPBERRY PI ZERO 2W

- A **Raspberry Pi** is a tiny, inexpensive computer
- Includes all the components of a standard PC (processor, USB ports, etc.)
- Runs various operating systems
- Can be used in many projects (desktop usage but also electronics, robotics, retro-gaming, and more)



RASPBERRY PI ZERO 2W COMPONENTS

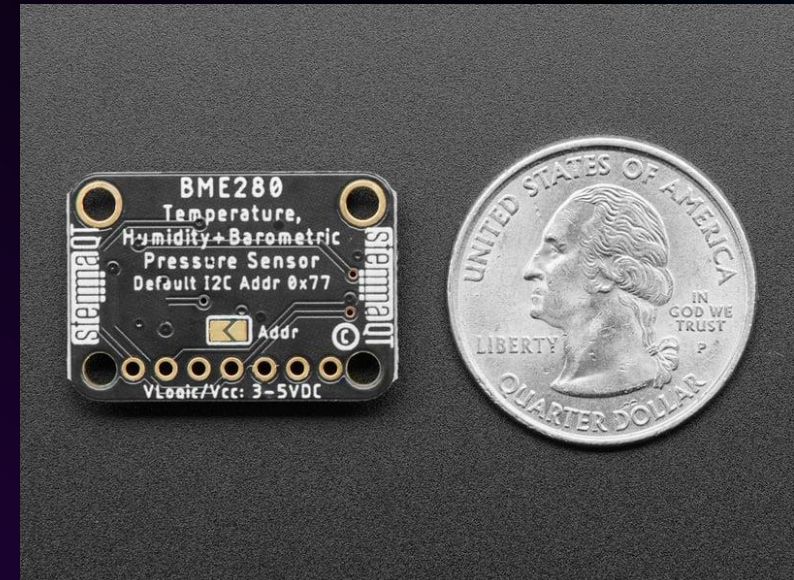
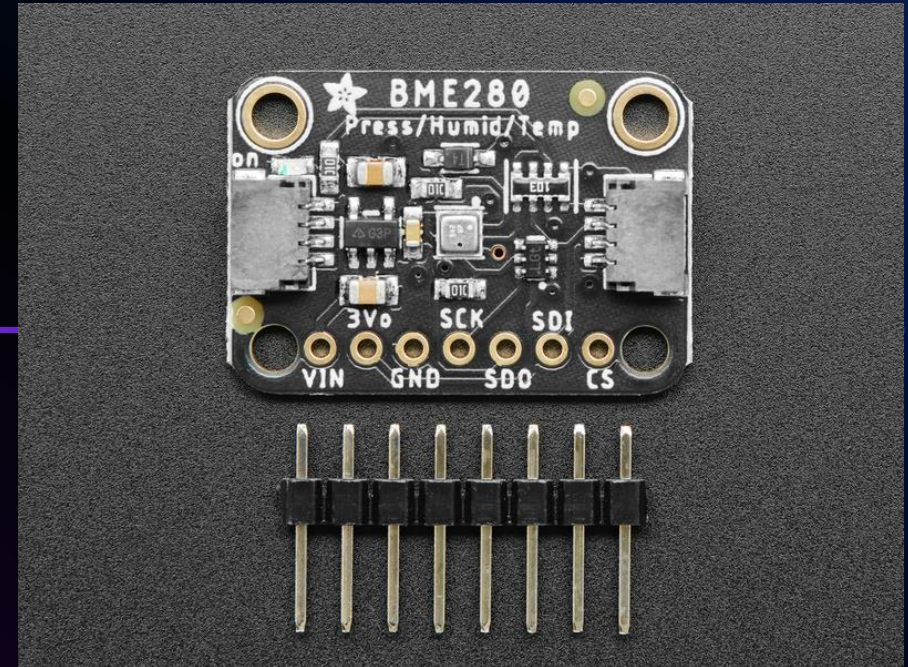
- Processor: Provides power for tasks like sensor data processing. (Quad-core 64-bit ARM Cortex-A53 processor)
- RAM: Efficiently handling of tasks IoT applications.
- Wireless Connectivity: Bluetooth and Wi-Fi for network connectivity and transmits sensor data
- GPIO: Connects to the sensor for collection of data
- Power Input: Powers the board.



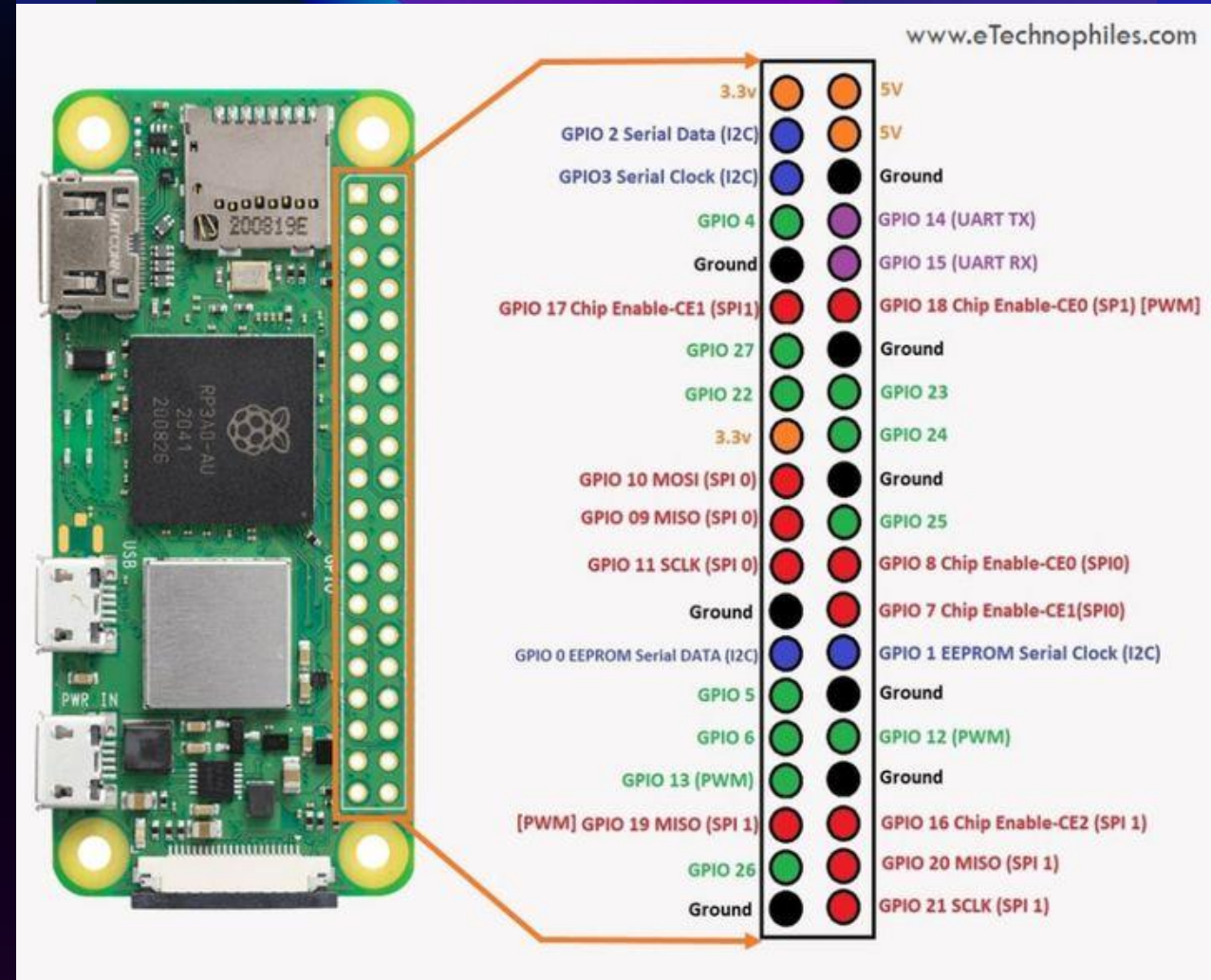
OUR SENSOR

BME 280 Humidity + Barometric Pressure + Temperature Sensor

- **Sensors** are devices that detect events or changes in the environment
- This is an environmental sensor that measures the humidity, barometric pressure and temperature with accuracy

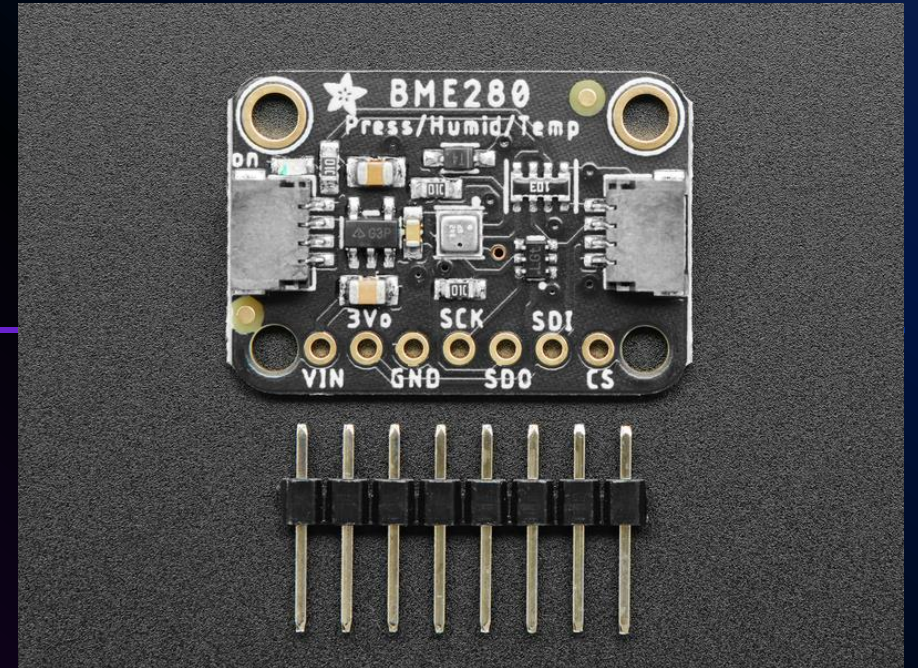


RPI ZERO 2W BOARD LAYOUT WITH PINOUTS



OUR SENSOR PINS

- VIN: Connects to the 3.3V pin to power the sensor
- GND: Connects to the ground pin
- SDA: Connects to the SDA pin for transmitting data
- SCL: Connects to the SCL pin for data transfer



ASSEMBLY

- We used a soldering kit to connect the sensor to the raspberry pi
- **Soldering wire** aka solder, is a fusible metal alloy used to join metal surfaces together
- When heated, solder melts then creating a strong electrical & mechanical bond when cooled



CODING

- Requires a virtual environment or venv as it is called in the code
- The packages are only allowed to be downloaded in the venv and it needs to be activated for the packages to be used in the code
- Its activated by source `~/venv/bin/activate`
- Packages read the data from the board and translated to readable text

```
AirQuality.py x Air.py x
1 import time
2 import board
3 import busio
4 from adafruit_bme280 import basic as adafruit_bme280
5
6 # Create I2C bus
7 i2c = busio.I2C(board.SCL, board.SDA)
8
9 # Create BME280 object
10 bme280 = adafruit_bme280.Adafruit_BME280_I2C(i2c, address=0x77)
11
12 def celsius_to_fahrenheit(celsius):
13     return (celsius * 9/5) + 32
14
15 while True:
16     try:
17         # Read sensor data
18         temperature_celsius = bme280.temperature
19         pressure = bme280.pressure
20         humidity = bme280.humidity
21
22         # Convert temperature to Fahrenheit
23         temperature_fahrenheit = celsius_to_fahrenheit(temperature_celsius)
24
25         # Print the readings
26         print("Temperature: {:.2f} C, {:.2f} F".format(temperature_celsius, temperature_fahrenheit))
27         print("Pressure: {:.2f} hPa".format(pressure))
28         print("Humidity: {:.2f} %".format(humidity))
29
30         # Wait for a few seconds before the next reading
31         time.sleep(2)
32
33     except KeyboardInterrupt:
34         print('Program stopped')
35         break
36
37     except Exception as e:
38         print('An unexpected error occurred:', str(e))
39         break
```



REAL WORLD APPLICATION

- This sensor can be used for measuring around plants to ensure that they are in the proper environments for growth, by checking the nearby temperature and humidity.

REAL WORLD APPLICATION



- **Sage**, (*Salvia officinalis*), aromatic herb of the mint family (Lamiaceae) cultivated for its pungent edible leaves.
- Native to the Mediterranean region and is used fresh or dried as a flavoring in many foods, particularly in stuffings for poultry and pork and in sausages.
- Ideal temperatures between 60-70° F are ideal for Sage.
- Ideal humidity is between 40% and 60%.

- **Chrysanthemums**, sometimes called mums or chrysanths are flowering plants
- Yellow chrysanthemums prefer cool temperatures ranging between 60-70° F
- Ideal humidity is between 40% and 60%



PROJECT WEBPAGE:

<https://omega-oxide.github.io/Air-Quality/>

THANK YOU
